Maya Arnott

New York, NY|(347) 409-0669|marnott276@gmail.com| www.linkedin.com/in/Maya-Arnott|Portfolio: https://maya-rae.github.io

Data Scientist with expertise in survival analysis, high-dimensional health data, and statistical modeling, with growing interest in real-world evidence and healthcare analytics. Recent M.S. graduate in Environmental Health Data Science from Columbia University Mailman School of Public Health.

EXPERIENCE

Data Analyst, *Columbia University Mailman School of Public Health* | Full-time | New York, NY March 2026 - Present

- Designing and applying mixture, survival, and regression models to identify exposure-disease associations across thousands of features in clinical and multi-omics data
- Building and maintaining an R package for high-dimensional data pipelines across metabolomic, genomic, and exposomic data
- Providing statistical consultation to collaborators, reviewing methodology and code to ensure accurate and reproducible results

Teaching Assistant, GIS Course, *Columbia University Mailman School of Public Health* |New York, NY July – September 2026

- Taught statistical and spatial analysis methods, including Bayesian statistics, to students with varying levels of technical background
- Designed assignments and datasets for lab work, supporting hands-on learning of statistical workflows

Life Sciences AI Research Expert, *Mercor Corporation* | Part-time Contractor | Remote May 2025 – Jan 2026

- Performed validation checks on AI-generated outputs, assessing factual accuracy and reasoning consistency in biology and life sciences; reduced hallucination rate by about 27% through iterative refinement with research teams

Research Technician II, *Duke University* | Full-time | Durham, NC July 2023 - June 2025

- Analyzed clinical trial data to evaluate SARS-CoV-2 and HIV-1 vaccine efficacy, synthesizing results into tables, figures, and reports for research strategy
- Automated data reporting pipelines in R to streamline experimental workflows
- Trained junior researchers on experimental procedures, such as BAMA and ELISA, and statistical interpretation from data reports

Air Quality Analyst, *Washington University in St. Louis*| Part-time | St. Louis, MO June 2022 - May 2023

- Developed scalable Python pipelines to process and clean global air quality datasets exceeding millions of records from the SPARTAN particulate monitoring network, improving analytical workflow efficiency by 40%
- Applied regression modeling and image analysis to quantify black carbon concentrations, supporting peer-reviewed environmental impact assessments

EDUCATION

M.S., Environmental Health Data Science *Columbia University* Mailman School of Public Health August 2025 - August 2026

Relevant Coursework: Biostatistical Methods (regression modeling, hypothesis testing, ANOVA), Data Science (data visualization & data wrangling), Public Health GIS (mapping), Bayesian Inference, Machine Learning for Epidemiology

Thesis, advised by Dr. Haotian Wu: "The environmental chemical exposome and prostate cancer severity: Race-stratified associations in a race-balanced cohort"

“ExpoMet: an R package for Exposome- and Metabolome-Wide Association Studies” – available locally on Github

B.A., Environmental Science *Washington University in St. Louis* August 2019 - May 2023

Major Environmental Biology **Minor** Urban Studies

SKILLS

- **Programming Languages:** SQL**,** R, Python, SAS
- **Libraries, Frameworks, Techniques:** tidyverse, Shiny, pandas, NumPy, scikit-learn, raster, XGBoost, matplotlib, tidycensus, terra, spdep
- **Databases & Data Handling:** PostgreSQL, MySQL, Excel, structured datasets, APIs
- **Tools:** Git, GitHub, VS Code, RStudio, Jupyter Notebook, Tableau, ArcGIS, Claude Code

PROJECTS

- Hypertension Prediction (NHANES): Applied LASSO, Random Forest, and XGBoost to national survey data; full ML pipeline with feature engineering and model validation
- Bangladesh Extreme Heat Analysis: Bayesian spatial modeling using INLA and EHSA to assess population-level health risk
- Texas Flood Risk Assessment: GIS and regression-based flood risk modeling using Python pipelines on large geospatial dataset e-cigarette aerosol samples